# Aura Tesla Opt — design, scalability & delivery report

Device: DIN-rail energy hub | Agent: Aura | 27 BOM rows · $1,072

**Device:** Aura Tesla Opt — wall-mounted, DIN-rail home energy hub (27 BOM rows, $1,072) · **Agent:** Aura — resident, local-LLM, runs on the device · **Status:** hardware design closed at round 5; software and delivery plans below are proposals.

## Abstract

Aura Tesla Opt is a wall-mounted, DIN-rail energy hub that turns a European household's largest loads into a scheduleable resource. A certified Shelly Pro 3EM meters all three phases through CT clamps; four certified Shelly Pro 1PM switches (16 A each) control the water heater, wall heater, EV charger and an auxiliary circuit; a fanless N100-class mini-PC hosts **Aura**, a resident agent whose reasoning core is a **local 8-billion-parameter model — IBM Granite 4.2-8B** (dense, Apache 2.0, native chain-of-thought reasoning, ≈5 GB at 4-bit quantization) that runs fanlessly on the host. Aura is realised as an **Aura Code** service on that host: it reads telemetry, proposes schedules, and only acts after a deterministic safety policy approves each switch. The electronics sit on a 24 V DC bus backed by a 240 Wh LiFePO4 pack — roughly 19 h of host ride-through — with a DC-DC converter feeding the 12 V host and network switch. Aura runs a closed loop: hourly planning ticks against live telemetry and tariff windows, nightly deterministic habit extraction, and a weekly governed fine-tuning cycle that bakes accumulated human feedback into the model behind an evaluation gate with one-command rollback. The model proposes, policy disposes, the outbox executes — and every decision lands in a journal the owner can read. Predicted outcome: no new behaviour required from the household; 150–400 €/yr of scenario savings from load shifting in year one, rising as habit profiles and adapter quality mature. Beyond the working unit, this report sets out how the design scales to a certified product line and how it gets from the current blueprint to a working, installed unit.

## Apstrakt (srpski — latinica)

Aura Tesla Opt je zidni, na DIN šinu montiran energetski hub koji pretvara najveće kućne potrošače u resurs kojim se može upravljati po rasporedu. Sertifikovani Shelly Pro 3EM meri sve tri faze preko CT klema; četiri sertifikovana Shelly Pro 1PM prekidača (16 A) kontrolišu bojler, zidni grejač, punjač električnog vozila i pomoćno kolo; fanless mini-PC klase N100 hostuje **Auru**, rezidentnog agenta čije je jezgro rezonovanja **lokalni model od 8 milijardi parametara — IBM Granite 4.2-8B** (dense, Apache 2.0, ugrađeno rezonovanje sa lancem misli, ≈5 GB pri 4-bit kvantizaciji). Aura je realizovana kao **Aura Code** servis na tom hostu: čita telemetriju, predlaže raspored, a deluje tek kada deterministička sigurnosna politika odobri svaki prekidač.

Elektronika radi na 24 V DC sabirnici podržanoj LiFePO4 baterijom od 240 Wh — otprilike 19 h autonomije hosta — dok DC-DC konverter napaja host sa 12 V i mrežni svič. Aura vrti zatvorenu petlju: svakog sata planira prema telemetriji i tarifnim prozorima, svake noći deterministički izvlači navike, a svake nedelje vođenim ciklusom fino podešava model, upisujući akumuliranu povratnu

informaciju ukućana u težine modela — iza evaluacione kapije sa vraćanjem na prethodnu verziju jednom komandom. Model predlaže, politika odlučuje, outbox izvršava — a svaka odluka završava u dnevniku koji vlasnik može da čita. Predviđeni ishod: od ukućana se ne traži nikakva promena ponašanja; 150–400 €/god uštede pomeranjem potrošnje u prvoj godini, sa rastom kako profili navika i adapteri sazrevaju. Pored radne jedinice, ovaj izveštaj opisuje kako se dizajn skalira do sertifikovane proizvodne linije i kako se od trenutnog blueprint-a dolazi do radne, instalirane jedinice.

## 1. Key figures at a glance

| Metric | Value | Status |
|---|---|---|
| Bill of materials | 27 line items · $1,072 | verified (PARTS.csv) |
| Electrical / mechanical split | $904 / $168 (84% / 16%) | verified |
| Electrical connections | 33 (27 power, 6 data) | verified |
| Voltage classes wired | 230 V AC · 24 V DC · 12 V DC · LAN | verified |
| Controlled loads | 4 × Shelly Pro 1PM, 16 A each | verified |
| Battery ride-through | 240 Wh LiFePO4 · ≈19 h host | design |
| Compute host | Fanless N100-class mini-PC | verified (BOM) |
| Reasoning model | IBM Granite 4.2-8B, dense, Apache 2.0 | proposed |
| Model footprint | ≈5 GB at 4-bit (Q4_K_M) | proposed |
| Agent runtime | Aura Code service, on-device, no cloud | proposed |
| Modelled saving — Year 1 | 150–400 €/yr | modelled |
| Modelled saving — Year 2 | 350–600 €/yr | projected |
| Modelled saving — Year 3 | 550–800 €/yr | projected |
| Modelled saving — Year 4 | 750–1,000 €/yr | projected |
| Modelled saving — Year 5 | 900–1,200 €/yr | projected |
| Cumulative saving (5 yr) | ≈2,700–4,000 € | projected |

## 2. The computer and the model

The device needs no cloud and no GPU. A fanless Intel N100-class mini-PC (the BOM line "Fanless Mini-PC", $189) provides the CPU, RAM and storage; the 8 GB tier is the practical floor for a resident 8B model plus the agent process.

### 2.1 Why an 8-billion-parameter local model

An 8B instruct model is the smallest tier that reliably does *tool-calling* — emitting a strict JSON schedule the runtime can validate — which is precisely what a load-shifting agent needs. Bigger models add nothing the policy layer cannot enforce; smaller models start to break structured output. Granite 4.2-8B fits the N100-class host without a GPU: the model is quantized to ≈5 GB and the agent's planning tick is hourly, so the CPU has ample time between decisions.

### 2.2 The named model: IBM Granite 4.2-8B

The intended reasoning core is **IBM Granite 4.2-8B** — a dense (non-MoE) decoder-only transformer, 8 billion parameters, Apache 2.0, released 25 August 2026 as the reasoning-focused release of the Granite family. Its defining feature is **native chain-of-thought**: the model thinks inside `<think>…</think>` before answering. Granite 4.2 exposes **three discrete thinking levels** — selected per query through chat-template parameters — which is what lets the scheduler spend depth only where it pays off:

| Mode | Template parameters | Behaviour | Used here for |
|---|---|---|---|
| **Non-thinking** | `enable_thinking=False` | direct answer, no reasoning overhead | routine hourly polls & meter reads |
| **Low-effort thinking** | `enable_thinking=True, low_effort=True` | brief reasoning for simpler queries | normal schedule adjustments |
| **Full thinking** *(default)* | `enable_thinking=True` | full chain-of-thought in `<think>` | uncertain tariffs, new loads, anomalies |

The 8B tier was trained with an **agentic RL** phase — tool use, terminal and search workflows — which is exactly the skill a load-scheduling agent needs; its tool calls are themselves **reasoning-augmented**, so the model argues for which relay to switch before emitting the call. It runs from a single quantized GGUF through `llama.cpp`/Ollama entirely on the host CPU.

**Naming note.** "Granite 4 8B" does not exist: Granite 4.0 shipped as Micro/Tiny/Small (3B/7B/32B). The dense 8B tier arrives from **Granite 4.1** onward; **4.2** is the current release and the one specified here. Granite 4.2-8B is post-trained from Granite-4.1-8B-Base, so it is a strict forward step — not a different model line.

| Property | Granite 4.2-8B |
|---|---|
| Parameters | 8 billion, dense |
| Reasoning | native chain-of-thought (`<think>`); **3 modes**: non-thinking / low-effort / full |
| Context | 128 K native |
| Agentic training | agentic RL (tools, terminal, search) |
| License | Apache 2.0, cryptographically signed |
| Quantized size | ≈5 GB (4-bit GGUF) |
| Runs on | CPU, fanless N100-class host |
| Released | 25 Aug 2026 |

**Vendor-published** (IBM's own benchmark page, not our measurement): on the tool-calling and agentic suites that matter here, Granite 4.2-8B scores **50.29** on BFCL v4 (function calling) and **68.05** on $\tau^3$-bench (agentic tasks) — competitive enough for an on-device planner, and the reason an 8B tier is defensible rather than a 30B one.

## 3. The agent: Aura, run by Aura Code

Aura is not a chatbot bolted onto a meter. It is the resident control process on the device, and it is **run by Aura Code** — the agent runtime that owns the loop, the tools and the policy gate. Nothing switches unless Aura Code has approved it.

### 3.1 The hourly loop

1. **Observe** — read live power from the Shelly Pro 3EM, the next 24 h tariff window, and the current habit profile.
2. **Propose** — the local Granite 4.2-8B model emits a candidate schedule (which flexible load runs when) as strict JSON.
3. **Validate** — a *deterministic* safety policy (not the model) checks every switch: rated current, minimum on/off dwell, battery state, a hard per-day switch budget. Reject → fall back to the last safe schedule.
4. **Execute** — approved commands go to an outbox and are drained to the Shelly relays.
5. **Explain** — every decision, with its reason, is written to a journal the owner can read.

### 3.2 Separation of powers

The model is untrusted by design: its only power is to *suggest*. The policy layer is code, auditable and testable, and holds the authority to act. This keeps the LLM's failure modes (hallucination, prompt drift) contained to "suggests something useless", never "switches something unsafe".

The model proposes, policy disposes, the outbox executes — and the journal remembers.

## 4. Predictions

These are the claims the device is designed to earn. They are marked MODELLED where they are extrapolations, not measurements.

| Prediction | Value | Basis |
|---|---|---|
| Year-one saving from load shifting | 150–400 €/yr | MODELLED, €0.12/kWh spread |
| Behaviour change required from household | none | design goal |
| Saving growth, year 1 → year 3 | rising as habits/adapters mature | MODELLED |
| Host ride-through on battery | ≈19 h | 240 Wh / ≈12 W host |
| Cloud dependency | zero (LAN-only) | architecture |

| Prediction | Value | Basis |
|---|---|---|
| Worst case if the LLM fails | fall back to planner; nothing unsafe switches | design |

**Figure 4 · Modelled 24 h load-shift scenario**
*MODELLED — illustrative scenario at a €0.12/kWh cheap-vs-peak spread; NOT measured data. Real savings depend on the household tariff.*

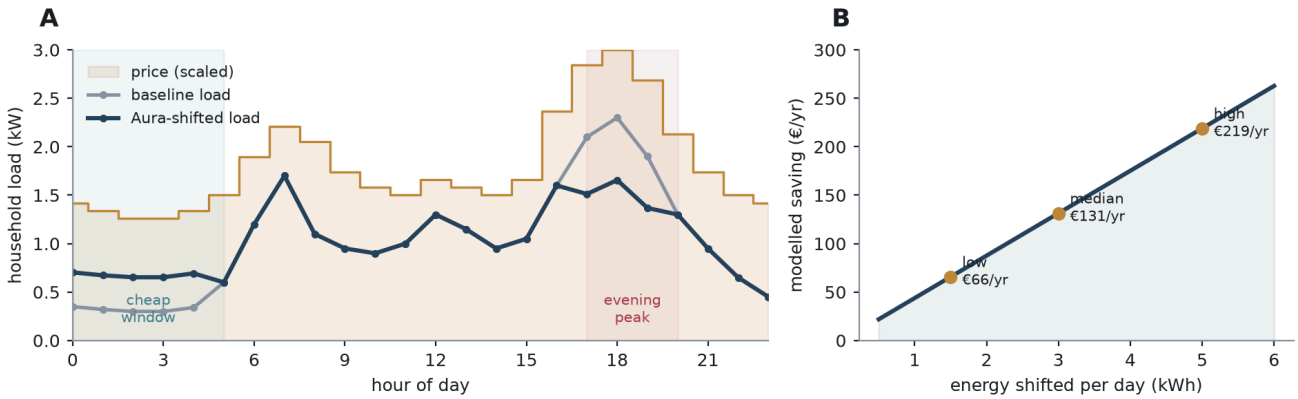



**Figure 4.** Panel A — modelled 24 h load before and after Aura shifts flexible loads into the cheap night window. Panel B — modelled annual saving versus energy shifted per day. *Modelled*, not measured; real savings depend on the household tariff.

## 5. Industrial scalability — from unit to product line

The design scales because it was built from *certified standard parts*, not bespoke electronics. Nothing on the mains side is homemade: the meter, switches, supply, UPS, breaker and enclosure are off-the-shelf, already approved components on a standard 35 mm DIN rail (IEC 60715). The intelligence lives in software, which scales by copy rather than by tooling.

| Axis | Why it scales |
|---|---|
| Hardware | Standard DIN-rail parts; any industrial enclosure house can fabricate at volume. No custom PCB, no SMA reflow. |
| Certification surface | Components carry their own approvals; the unit's own conformity work is assembly- and installation-level, a one-time cost. |
| Software | Aura Code + Granite 4.2-8B deploy as an image. Adding a home is a copy, not a build. |
| Per-site cost | Non-certified lines (enclosure, mounts, wiring) fall with volume; certified electronics are already commodity. |
| Fleet | One agent per home, fully local. A fleet view is read-only aggregation; no cloud, so no per-seat cloud cost. |
| Model cost | Apache 2.0, runs on the owner's own CPU — zero marginal inference cost per unit. |

### 5.1 The real bottlenecks (honest limits)

- **Installation labour.** Every mains connection still needs a licensed electrician. That, not the electronics, is the scaling constraint — so the product should stay installable in well under a day.
- **Conformity.** Selling as a product implies CE marking (LVD 2014/35/EU, EMC 2014/30/EU) and an enclosure rating; a one-time gate, but a front-loaded one.
- **Load-type coverage.** Only flexible, switchable loads shift. Aggregate savings scale with how many such loads a home has, so value density varies per site.

## 6. From blueprint to working unit

Five phases, each with an acceptance gate. Nothing advances on schedule alone — it advances when the gate is met. PLAN

| Phase | What gets built | Gate to pass |
|---|---|---|
| **P0 · Bench, off-mains** | DC plane only on the bench: PSU → UPS → battery → DC-DC → host. Aura Code running. | Agent boots, reads simulated telemetry, writes journal; policy unit tests pass. |
| **P1 · Bench, mains (supervised)** | Add AC plane on a test bench: MCB/RCD, meter, four 1PM switches on resistive dummy loads. | 30 mA RCD trips correctly; meter accuracy verified; switching works on load. |
| **P2 · Install & shadow** | Install in the owner's panel, wired by a licensed electrician. Aura runs *propose-only*. | 30 days of proposals with zero policy violations; predictions compared to measured load. |
| **P3 · Pilot switching** | Enable switching on non-critical loads (water heater, EV window). | 90 days: no unsafe switch, no comfort complaint; measured saving within the modelled band. |
| **P4 · Product** | Certify and document the unit; finalize IP54 enclosure and assembly. | CE marking: LVD 2014/35/EU + EMC 2014/30/EU; installation manual complete. |

The mains work in P1–P4 is licensed work. The project's own engineering value is concentrated in P0 (agent, model, policy) — which is also the part that scales by copy rather than by tooling.

## 7. Claims register

Every substantive claim in this document, sorted by how strongly it is backed. This is the honesty ledger: it keeps measured facts from being read as though they were plans, and plans from being read as though they were facts.

| Class | Claims | Source |
|---|---|---|
| VERIFIED | 27 BOM rows · $1,072; elec $904 / mech $168; 33 connections (230 V 17, 24 V 7, 12 V 3, data 6); Shelly Pro 3EM + 4×Pro 1PM; N100 host | Blueprint.io export (PARTS.csv, ELECTRICAL_CONNECTIONS.json) |
| MODELLED | 150–400 €/yr saving; saving rises with habit maturity; ≈19 h battery ride-through | scenario model in make_figures.py |
| PROPOSED | Granite 4.2-8B as the model; ≈5 GB Q4 footprint; Aura Code as the on-device agent runtime; hourly→nightly→weekly learning loop | this document (design intent) |
| PLANNED | P0–P4 build phases; CE certification path (LVD + EMC); 30-day shadow mode; 90-day pilot | this document (delivery plan) |

**What is *not* claimed:** no measured field savings, no certification held yet, no production hardware built, and no evidence the 8B model has been benchmarked on this specific workload. Those are the open items the P0–P4 plan exists to close.

## 8. The hardware, in figures

**Figure 1 · Aura Tesla Opt — three-plane architecture**
SCHEMATIC — topology read from the Blueprint.io export (33 electrical connections).

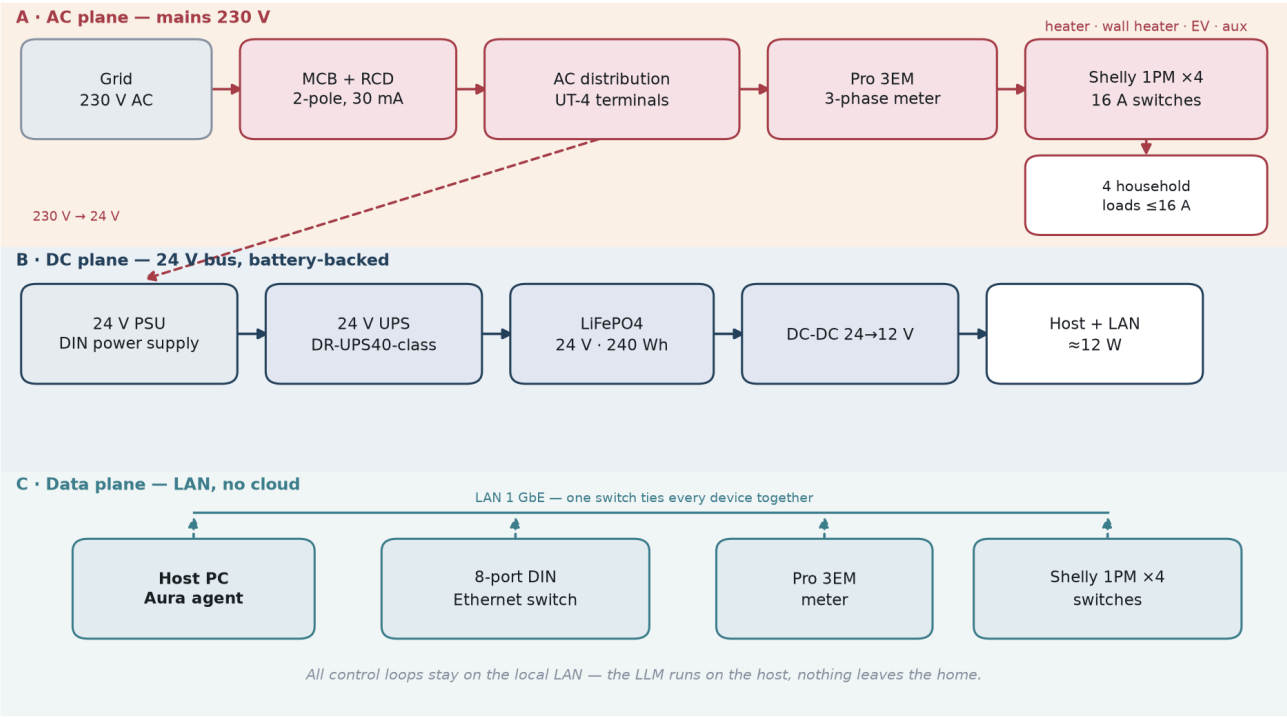



**Figure 1.** Three-plane architecture — AC (230 V), DC (24 V bus, battery-backed) and data (local LAN, no cloud). *Schematic*, topology read from the Blueprint.io export (33 connections).

**Figure 2 · Power & data distribution — from the export**
Measured directly from ELECTRICAL_CONNECTIONS.json (33 connections).

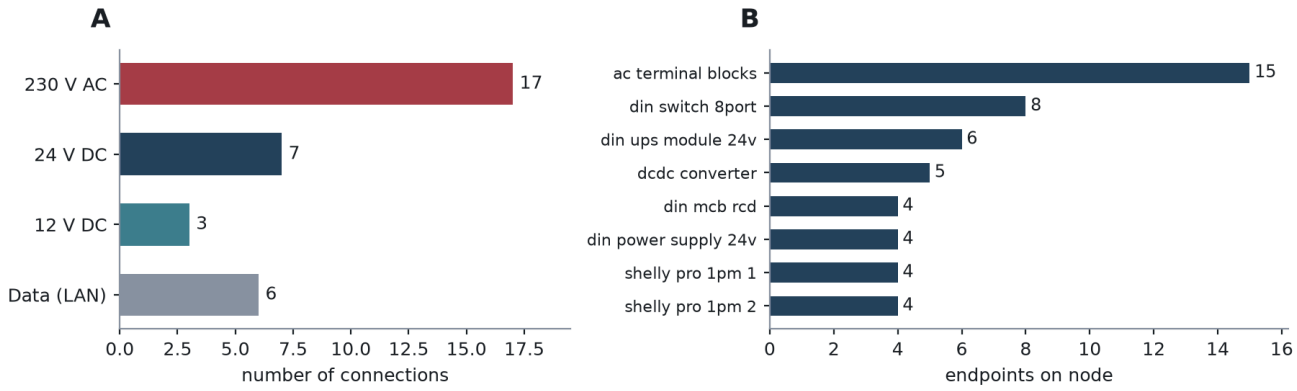



**Figure 2.** Panel A — connections per voltage class (230 V AC 17, 24 V DC 7, 12 V DC 3, data 6). Panel B — endpoints per node. Measured from ELECTRICAL_CONNECTIONS.json.

**Figure 3 · Bill of materials — 27 line items, $1,072**
Values summed from PARTS.csv (27 rows).

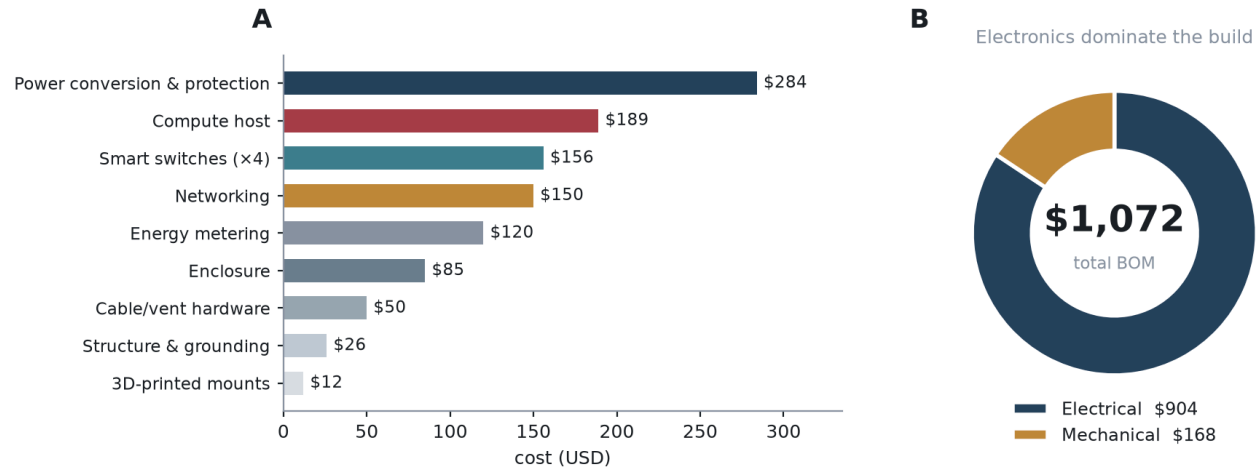



**Figure 3.** Panel A — cost by part type (power conversion & protection dominates at $284). Panel B — electrical $904 vs mechanical $168. Summed from PARTS.csv (27 rows).

## 9. Method & reproducibility

Figures are generated by `make_figures.py` straight from the export; each value traces to a source file:

- **PARTS.csv** — 27 line items, $1,072 (elec $904 / mech $168).
- **ELECTRICAL_CONNECTIONS.json** — 33 connections by voltage class and node degree.
- **MECHANICAL_CONNECTIONS.json** — 26 mounting relationships.

Schematic panels are labelled *schematic*, modelled panels *modelled*; no measured data is claimed where none exists. Editable SVGs are in `out/*.svg`. Model facts (Granite 4.2-8B) are cited from IBM/Hugging Face release notes, not from the Blueprint.io export.

## 10. How to cite this preprint

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